TrES-1b

R-1724

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-1_210802` · created 2026-08-02T16:54:18+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459428.7465 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.173 +/- 0.014
Transit depth (Rp/Rs)^2	2.99 +/- 0.48 [%]
Orbital Inclination (inc)	85.3 +/- 0.25
Airmass coefficient 1 (a1)	1.159 +/- 0.011
Airmass coefficient 2 (a2)	-0.135 +/- 0.033
Scatter in the residuals of the lightcurve fit is	0.93 %
Best Comparison Star	None
Optimal Aperture	4.88
Optimal Annulus	12.77
Transit Duration (day)	0.073 +/- 0.0051

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.173 ± 0.014 vs 0.1358 (deviation 2.55σ)
Duration fit vs published	0.073 d vs 0.1042 d (deviation 5.86σ)
Mid-time O–C	-6.5 min
Depth SNR	11.9
Residual scatter	0.93 %

Light curve

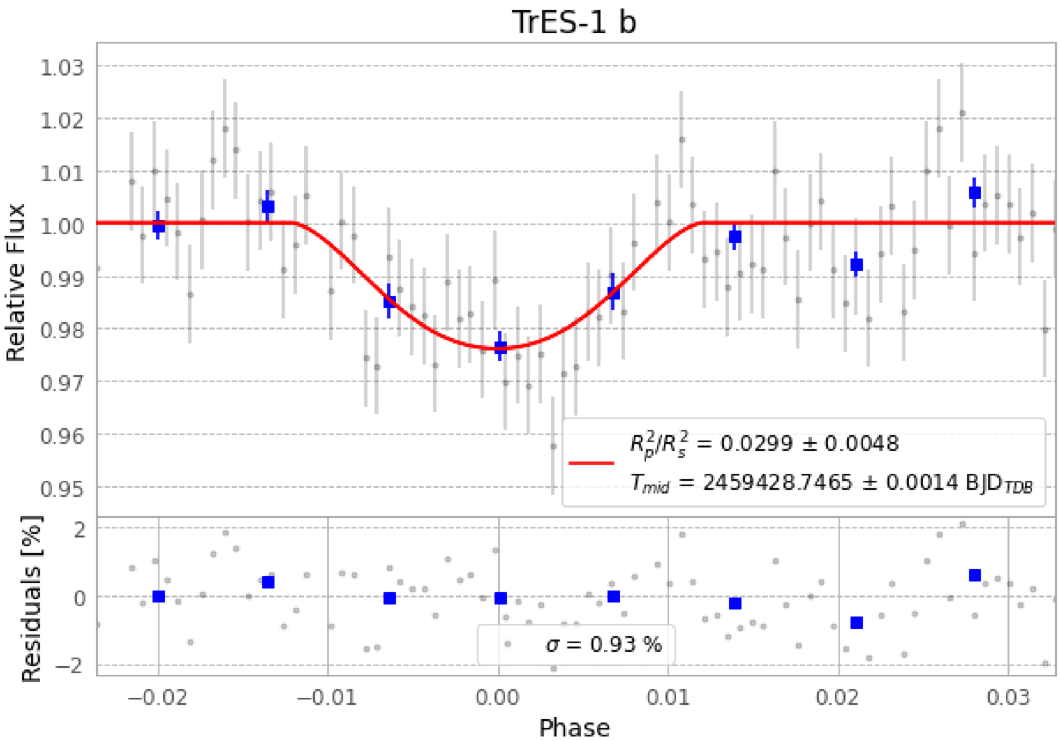

Figure 1 — FinalLightCurve_TrES-1 b_2021-08-01.png (SHA-256 84d1da22e5aeadde…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [355, 196], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 a43a3659735626ae…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

81 FITS frames (53 MB), `TRES-1210802040612.FITS` → `TRES-1210802081212.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-210802.log (cd7ed8009507…), FinalLightCurve_TrES-1 b_2021-08-01.png (84d1da22e5ae…), FinalLightCurve_TrES-1 b_2021-08-01.csv (2e8b3a48d44c…)

Compute resources

Wall time 245.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)